

Week 2: Authentication, Authorization and Incident Management

Authentication System

The second week focused on implementing secure user authentication.

The system uses **JWT-based authentication** along with password hashing using **bcrypt**.

Three different user roles were implemented:

```text

Reporter

Admin

Rescue Team

Each role has different responsibilities within the application:

- **Reporter:** Submit and view their own incidents.
- **Admin:** View all incidents and assign rescue teams.
- **Rescue Team:** View assigned incidents and update their status.

## Role-Based Access Control

A role middleware was introduced to ensure that users can only access operations permitted for their role.

For example:

```text

Reporter → Submit Incident

Admin → Assign Rescue Team

Rescue Team → Update Incident Status

This prevents unauthorized users from accessing administrative or rescue-team functionality.

Incident Management

The incident management system was then implemented.

An incident can contain:

```text

Message

Reporter

Location

Category

Confidence

Urgency

Priority

Status

Assigned Rescue Team

Notes

Status History

Location information includes:

```
```text
Latitude
Longitude
Address
```

The incident lifecycle was designed to track the progress of a disaster report from submission to resolution.

Incident Status

The incident can move through different stages:

```
```text
```

pending

↓

assigned

↓

accepted

↓

en\_route

↓

on\_site

↓

resolved

Status history is maintained with timestamps so that the progress of each incident can be tracked.

## Outcome

By the end of Week 2, authentication, authorization and the basic incident-management workflow had been